

Europass Curriculum Vitae



Personal information

Name / Surname

Home page

Nationality

Date of birth

Mason Emanuele

<http://www.lordmzn.it>

Italian

9 July 1988



Work experience

4th Aug 2021 - present

Main tasks and responsibilities

Digital solutions manager

I own the technical direction of EMS, ENERSEM's cloud platform for energy monitoring and optimization, which drives more than a third of company revenue. EMS collects time-series data from 30+ industrial and commercial sites across 15+ clients, with continuous storage since 2020. I designed and built the core time-series analysis library — over 30 composable analysis functions with a no-code interface — and the remote setpoint optimization system running on plants worth 50k€+. Since 2021 I have been coordinating 1.5 FTE of external developers; for the past year and a half I have also been leading two internal colleagues. I supervise funded research projects and contribute to grant writing.

ENERSEM s.r.l.

Name and address of employer

9th Jul 2020 - 3rd Aug 2021

Main tasks and responsibilities

Socio lavoratore

Formal entry into ENERSEM as an equity partner. Took ownership of EMS development, transitioning from external research consultant to internal platform builder responsible for the product's technical direction.

ENERSEM s.r.l.

Name and address of employer

6th Mar 2018 - 8th Jul 2020

Main tasks and responsibilities

Consultant for ENERSEM

Developed energy models for industrial plants, offices and commercial buildings, combining data analysis, machine learning and model predictive control to extract insights on energy use and efficiency. Led the technical work on two funded research projects: TEPORE (2018–2019), developing an MPC algorithm for smart thermostats and occupancy-based control; and LombHe@t (2020–), integrating heat pump modelling into control algorithms and analysing district heating consumption profiles.

ENERSEM s.r.l.

Name and address of employer

2nd Feb 2018 - 3rd Apr 2018

Main tasks and responsibilities

Modelling and managing an alpine hydroelectric plant

As a private consultant, I developed and applied algorithms for modelling the historical operations of an hydroelectric reservoir (Cancano lake, North of Italy) in multiple periods, with particular focus on evolving hydroelectric markets. The research aimed at quantifying the adaptive response of system management, and at improving the assessment of system performance. To this end, we applied algorithms of Inverse Reinforcement Learning to a lumped, conceptual model of the hydroelectric plant.

Prof. Andrea Castelletti, Politecnico di Milano

Name and address of employer

1st Feb 2017 - 15th Apr 2017

Desk study River Restoration wiki

Main tasks and responsibilities	Analysis of benefits provided by implemented river restoration schemes across Europe through a review of existing databases, including the LIFE+ RESTORE wiki and the FP7 REFORM wiki, in order to provide evidence of river restoration works that address flood risk and ecological status. In collaboration with CIRF, Centro Italiano per la Riqualificazione Fluviale.
Name and address of employer	Wetlands International - European Association
1st Mar 2015 - 31st Jun 2017	
Main tasks and responsibilities	Collaborator in teaching activities Class of "Modellistica e simulazione" in Italian A.Y. 2015/2016 and 2016/2017: Excel exercises of discrete system simulation, discretization of continuous systems, linear and non linear programming using Excel solver. 20 hours of computer lab. Class of "Modelling and Simulation" A.Y. 2015/2016: non linear systems in continuous time, stability and canonical responses, control and observability of linear systems. 12 hours of class exercises. Class of "Analisi dei Sistemi" in Italian A.Y. 2014/2015: linear systems in continuous and discrete time, non linear systems, stability and canonical responses. 8 hours of class exercises and 4 hours of computer lab.
Name and address of employer	Prof. Giorgio Guariso, Politecnico di Milano. Prof. Alessandra Gragnani, Politecnico di Milano.
1 Oct 2013 - 30 Sep 2014	
Main tasks and responsibilities	Research fellow at Politecnico di Milano. Design and coding of a simulation software for dynamic models and application to the regulation of Red-Thai Binh hydroelectric basin in Vietnam.
Name and address of employer	Prof. Rodolfo Soncini-Sessa, Politecnico di Milano.
Education and training	
Oct 2014 - 17 Jan 2018	
Title	PhD in Information Technology, <i>cum laude</i>
Principal subject	Beyond full rationality: modelling tradeoff dynamics in multiobjective water management
Skills acquired	We have been researching how to model the behavior of human operators in water systems with multiple targets. We expand the traditional normative approaches based on the assumption of a utility maximizer into a multi objective context, where multiple, equivalently optimal behaviors co-exist. We develop a non-parametric approach based on inverse reinforcement learning to identify the multiple utility functions of an operator, and a parametric agent based negotiation protocol to identify the operator's tradeoff among multiple utilities and the tradeoff evolution in time. The methodologies are demonstrated on both synthetic case studies and real settings. Management of natural resources, especially water; control theory: control of hybrid systems; machine learning: supervised learning and reinforcement learning; multi-agent systems; game theory; multi criteria decision aiding; epistemology of research in social sciences; experimental methods in engineering.
Supervisor and school	Prof Andrea Castelletti and Matteo Giuliani, PhD, Department of Engineering, Information, Bioengineering, Politecnico di Milano
25 May 2014 - 1 Jun 2014	
Principal subjects / occupational skills learned	Spring school in <i>Multi Criteria Decision Aiding</i> , Perugia 2014
Name and kind of school	Aiding the <i>multi criteria</i> decision: multi attribute value theory, ELECTRE and PROMETHEE methods, analytical hierarchical process.
	Università degli studi di Perugia
Oct 2010 - May 2013	
Grade	Master of Science (M.Sc) in Environmental and Land Planning Engineering: Natural Resources Planning and Management
Principal subjects / occupational skills learned	110/110 <i>cum laude</i>
Name and kind of school	Management of natural resources, numeric modeling of fluvial systems, multi-objective optimization, Environmental impact assessment, land planning.
	Politecnico di Milano
Oct 2012 - Dec 2012	
	Visiting scholar at PennState

Main tasks	Visiting scholar at Hydroinformatics Research Group - Department of Civil and Environmental Engineering. Pennsylvania State University, USA.
Principal subjects / occupational skills learned	Use of multi objective evolutionary algorithm for complex mathematical programming (MOEAframework and BorgMOEA).
Oct 2007 - Sep 2010	Bachelor of Engineering (B. Eng) in Environmental and Land Planning Engineering
Grade	102/110
Principal subjects	Core engineering subjects: mathematics, physics, chemistry, informatics. Insights of economy, building science, geology, hydrology, environmental chemistry.
Name and kind of school	Politecnico di Milano
Sep 2003 - Jul 2007	Italian high school "liceo scientifico"
Grade	92/100
Principal subjects	General high level education on italian, latin and english literature; understanding of written latin; speaking and writing modern english; italian writing of short essays and newspaper articles; overview of main historical and philosophical topics; art history; basic knowledge of mathematics, physics, chemistry, biology, geology, astronomy, information technology, technical and artistic drawing.
Name and kind of school	Liceo Scientifico A. Tosi, Busto Arsizio
Publications	
Journal	
[A5]	E. Mason, M. Giuliani, A. Castelletti, and F. Amigoni. Identifying and modeling dynamic preference evolution in multipurpose water resources systems. <i>Water Resources Research</i> , 54(4):3162–3175, 2018
[A4]	S. Bizzi, A. Cominola, E. Mason, A. Castelletti, and K. Paik. Multicriteria optimization model to generate on-dem optimal channel networks. <i>Water Resources Research</i> , 54(8):5727–5740, 2018
[A3]	M. Giuliani, Y. Li, A. Cominola, S. Denaro, E. Mason, and A. Castelletti. A Matlab toolbox for designing Multi-Objective Optimal Operations of water reservoir systems. <i>Environmental Modelling & Software</i> , 85:293–298, 2016
[A2]	F. Amigoni, A. Castelletti, P. Gazzotti, M. Giuliani, and E. Mason. Using Multiagent Negotiation to Model Water Resources Systems Operations. In <i>International Conference on Autonomous Agents and Multiagent Systems</i> , pages 51–72. Springer, 2016
[A1]	M. Giuliani, A. Castelletti, F. Pianosi, E. Mason, and P.M. Reed. Curses, tradeoffs, and scalable management: advancing evolutionary multi-objective direct policy search to improve water reservoir operations. <i>Journal of Water Resources Planning and Management</i> , 142(2):04015050, 2015
Conference proceedings	
[B11]	E. Mason, S. Bizzi, A. Cominola, A. Castelletti, and K. Paik. Exploring 3D optimal channel networks by multiple organizing principles. In <i>EGU General Assembly</i> , Vienna (Austria), 2017

- [B10] M. Giuliani, Y. Li, A. Cominola, S. Denaro, A. Castelletti, and R. Soncini Sessa.
M3O: a Matlab toolbox for designing Multi-Objective Optimal Operations of water reservoir systems.
In *8th International Congress on Environmental Modelling and Software (iEMSs 2016)*, Toulouse, FR, 2016., Toulouse (FR), 2016
- [B9] F. Amigoni, A. Castelletti, P. Gazzotti, M. Giuliani, and E. Mason.
Water Resources Systems Operations via Multiagent Negotiation.
In *Proceedings of the 2016 International Conference on Autonomous Agents & Multi-agent Systems*, pages 1379–1380, Singapore, 9-13 May 2016. International Foundation for Autonomous Agents and Multiagent Systems
- [B8] E. Mason, M. Giuliani, A. Castelletti, and F. Amigoni.
Modelling water operators' time varying risk aversion by an agent-based multi lateral negotiation protocol.
In *World Environmental & Water Resources Congress (ASCE EWRI 2016)*, West Palm Beach (Florida), 2016
- [B7] E. Mason, P. Gazzotti, F. Amigoni, M. Giuliani, and A. Castelletti.
Modelling Tradeoffs Evolution in Multipurpose Water Systems Operation in Response to Extreme Events.
In *2015 AGU Fall Meeting*, San Francisco (California), 14-18 December 2015
- [B6] M. Giuliani, A. Castelletti, F. Pianosi, E. Mason, and P.M. Reed.
Curses, tradeoffs, and scalable management: advancing evolutionary multi-objective direct policy search to improve water reservoir operations.
In *World Environmental & Water Resources Congress (ASCE EWRI 2015)*, Austin (Texas), 17-21 May 2015
- [B5] M. Giuliani, D. Anghileri, A. Castelletti, E. Mason, M. Micotti, R. Soncini-Sessa, and E. Weber.
Integrated and Sustainable Water Management of Red–Thai Binh Rivers System Under Change.
In *2014 AGU Fall Meeting*, San Francisco (California), 15-19 December 2014
- [B4] M. Giuliani, E. Mason, A. Castelletti, F. Pianosi, and R. Soncini-Sessa.
Universal approximators for direct policy search in multi-purpose water reservoir management: A comparative analysis.
In *Proceedings of the 19th IFAC World Congress*, Cape Town (South Africa), 24-29 August 2014
- [B3] M. Giuliani, E. Mason, A. Castelletti, and F. Pianosi.
Universal approximators for multi-objective direct policy search in water reservoir management problems: a comparative analysis.
In *EGU General Assembly*, Vienna (Austria), 27 April - 2 May 2014
- [B2] M. Micotti, F. Pianosi, S. Bizzi, E. Mason, and E. Weber.
Analyzing tradeoffs between hydropower production and hydrological alteration to support water resources planning in large river systems.
In *EGU General Assembly*, Vienna (Austria), 27 April - 2 May 2014
- [B1] E. Mason, A. Cominola, A. Castelletti, S. Bizzi, and K. Paik.
A multi-objective optimization framework to model 3d river and landscape evolution processes.
In *EGU General Assembly*, Vienna (Austria), 07 - 12 April 2013

Awards

Apr 2016

From

IDEAS 2016 Best Paper Award – [B10]

IDEAS Organizing committee and AAMAS Workshop Chairs

Jul 2012

From

Winner of grant “Tesi all'estero”

Politecnico di Milano

Personal skills

Languages

Italian (mother tongue). Proficient English. Basic French.

Mother tongue

Italian

Other languages

*Self-assessment
European level^(*)*

English

French

Understanding		Speaking		Writing
Listening	Reading	Spoken interaction	Spoken production	
C1 Proficient user	C1 Proficient user	B2 Independent user	B2 Independent user	C1 Proficient user
B1 Independent user	B1 Independent user	A1 Basic user	A1 Basic user	A1 Basic user

^(*) Common European Framework of Reference (CEF) level

Communication skills

Documentation and reports – Manual of DMMT software, used now by Vietnamese water agency under the project IMRR. Final scientific report of the studies performed (Deliverable D7.1 of IMRR)

Reports of work – MODUS workshop at Nizza University, Sophia Antipolis, with a presentation titled “Concepts behind models”, showing the work developed during the course of “Modellistica e Governo del Territorio” (Modelling and land management), Jan 2012.

Writing of scientific papers – See section *Publications* for the articles.

Organizational /
managerial skills

People management – Coordination of developers external to Enersem. Management of internal reporting.

Conflict resolution – Member of “Consiglio dei Garanti” of Partito Democratico of Varese province, the internal judicial board, Oct 2013 - Dec 2016.

Prioritization, decision making and team management – Part of the secretariat of local branch of Giovani Democratici - the youth branch of the Partito Democratico, Jun 2013 - Jan 2015.

Computing Skills

Cloud/Platform

Data/ML

Scientific computing

Systems/infra

Web

Python (Flask), Javascript (React), PostgreSQL, Kubernetes, Google Cloud

Jupyter notebooks in Python with pandas, sklearn and statsmodels

Matlab, R

Bash, C++, Elixir, Java, Git

PyDMMT library: <https://github.com/Lordmzn/pydmmmt>

basic knowledge of html, CSS, Javascript

Small website design: www.cnai.info, www.lordmzn.it

Latex, Apple suite, MS Office

OSX; Linux Ubuntu; Windows

Personal productivity tools

Operating systems

Driving license

B level driving license.

Personal interests

I have been practicing martial arts since 2010, with long experience in traditional Chinese martial arts and more recently, in European historical fencing. I am a black belt in Tai Chi and Shaolin Chang style at the European Federation of Kung fu Schools (FESK) and a member of the 1595 historical fencing club. I am also a fan of Formula 1. In my spare time, I hike in the mountains and go to rock concerts.

Milan, June 5, 2026,

I authorize the processing of the personal data contained in my curriculum vitae according to art. 13 D. Lgs. 196/2003 (available at <http://www.garanteprivacy.it/garante/doc.jsp?ID=1311248>) and to art. 13 GDPR 679/16.

Firma:
